Linear Algebraic Solvers

Mathematical Formulation of the Linear System

In the context of implicit finite difference discretizations for transient partial differential equations (such as the 1D diffusion equation), the advancement from time level n to $n+1$ requires solving a coupled system of algebraic equations across all interior grid points simultaneously. In general algebraic form, this linear system is expressed as:

$$\mathbf{A}\mathbf{x}^{n+1} = \mathbf{b}^n \tag{1}$$

where:

- $\mathbf{A} \in \mathbb{R}^{N\times N}$ is the square coefficient matrix of size $N \times N$,
- $\mathbf{x}^{n+1} \in \mathbb{R}^N$ is the vector of unknown field variables at time level $n+1$ (i.e., $\mathbf{u}^{n+1}$),
- $\mathbf{b}^n \in \mathbb{R}^N$ is the right-hand side (RHS) load vector containing the known solution from time level n along with prescribed boundary conditions.

In expanded matrix-vector notation, Equation (1) is represented as:

$$\begin{bmatrix} a_{0,0} & a_{0,1} & a_{0,2} & \cdots & a_{0,N-1} \\ a_{1,0} & a_{1,1} & a_{1,2} & \cdots & a_{1,N-1} \\ a_{2,0} & a_{2,1} & a_{2,2} & \cdots & a_{2,N-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{N-1,0} & a_{N-1,1} & a_{N-1,2} & \cdots & a_{N-1,N-1} \end{bmatrix} \begin{bmatrix} x_0^{n+1} \\ x_1^{n+1} \\ x_2^{n+1} \\ \vdots \\ x_{N-1}^{n+1} \end{bmatrix} = \begin{bmatrix} b_0^n \\ b_1^n \\ b_2^n \\ \vdots \\ b_{N-1}^n \end{bmatrix} \tag{2}$$

Solving Equation (1) accurately and efficiently is central to the execution of any implicit scheme.

From a general linear algebra perspective, treating the coefficient matrix $\mathbf{A}$ in its full algebraic representation necessitates the deployment of general direct solvers, such as **Gaussian Elimination**, which will be discussed in detail in subsequent sections. However, applying standard Gaussian Elimination to a dense $N \times N$ system incurs an asymptotic computational complexity of $\mathcal{O}(N^3)$ operations and requires $\mathcal{O}(N^2)$ storage.

Upon closer inspection of the governing discrete algebraic equations, the tridiagonal structure of the coefficient matrix $\mathbf{A}$ is mathematically inevitable. The spatial second-derivative operator is approximated using a compact three-point central difference stencil, which inherently couples the state variable at node j exclusively to its immediate spatial neighbors, $j-1$ and $j+1$. Because no long-range node interactions exist in the local differential stencil, the off-diagonal entries vanish identically beyond the first sub- and super-diagonals.

Consequently, the resulting system can be solved straightforwardly and efficiently using the **Thomas Algorithm**(Tridiagonal Matrix Algorithm, TDMA)—a streamlined variant of Gaussian Elimination that eliminates unnecessary operations on zero entries. This algorithm resolves the system in two simple passes (a forward elimination sweep followed by a backward substitution sweep), executing in $\mathcal{O}(N)$ computational time with $\mathcal{O}(N)$ memory storage. Furthermore, because the coefficient matrix generated by this implicit scheme is strictly diagonally dominant under standard conditions, the Thomas Algorithm is guaranteed to be unconditionally stable without requiring numerical pivoting.